

Packet Tracer – DHCP Relay Across Multiple VLANs and Subnets

Objectives

Part 1: Build a Multi-VLAN Topology

Part 2: Configure VLANs and Trunking

Part 3: Configure Router-on-a-Stick Inter-VLAN Routing

Part 4: Configure a Centralized DHCP Server

Part 5: Configure DHCP Relay (IP Helper Address)

Part 6: Verify DHCP Address Assignment

Part 7: Verify Inter-VLAN Connectivity

Background / Scenario

A company has multiple departments separated into different VLANs.

Instead of deploying a DHCP server in every VLAN, the company wants a single centralized DHCP server located in the Server VLAN.

Because DHCP broadcasts do not cross routers, you must configure DHCP Relay (IP Helper Address) so that clients in other VLANs can obtain IP addresses from the centralized server.

In this lab, you will:

- Create multiple VLANs
- Configure VLAN trunking
- Configure Router-on-a-Stick
- Configure a centralized DHCP server
- Configure DHCP Relay
- Verify that clients in different VLANs receive addresses automatically
- Verify communication between VLANs

This is a common enterprise network design used in schools, offices, hospitals, and large organizations.

Instructions

Part 1: Build the Topology

Step 1: Add Devices

Add:

- 1 Router (2911)
- 1 Switch (2960)
- 1 Server
- 2 PCs

Rename:

- Router0 → **R1**
 - Switch0 → **SW1**
 - Server0 → **DHCP-SERVER**
 - PC0 → **PC-A**
 - PC1 → **PC-B**
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Step 2: Connect Devices

Use Copper Straight-Through cables.

Device	Interface	Device	Interface
PC-A	Fa0	SW1	Fa0/1
PC-B	Fa0	SW1	Fa0/2
DHCP-SERVER	Fa0	SW1	Fa0/3
SW1	Fa0/24	R1	G0/0

Wait for all links to turn green.

Part 2: Configure VLANs

We will create:

VLAN Name	Network
10 SALES	192.168.10.0/24
20 HR	192.168.20.0/24
99 SERVER	192.168.99.0/24

Configure SW1

```
enable
configure terminal
```

```
vlan 10
name SALES
```

```
vlan 20
name HR
```

```
vlan 99
name SERVER
```

```
exit
```

Assign Access Ports

```
interface fa0/1
switchport mode access
switchport access vlan 10
```

```
interface fa0/2
switchport mode access
switchport access vlan 20
```

```
interface fa0/3
```

```
switchport mode access
switchport access vlan 99
```

Configure Trunk Port

```
interface fa0/24
switchport mode trunk

end
```

Part 3: Configure Router-on-a-Stick

Configure Subinterfaces

```
enable
configure terminal
```

```
interface g0/0
no shutdown
```

VLAN 10 Gateway

```
interface g0/0.10
encapsulation dot1Q 10

ip address 192.168.10.1 255.255.255.0

ip helper-address 192.168.99.10
```

VLAN 20 Gateway

```
interface g0/0.20
encapsulation dot1Q 20

ip address 192.168.20.1 255.255.255.0

ip helper-address 192.168.99.10
```

VLAN 99 Gateway

```
interface g0/0.99
encapsulation dot1Q 99

ip address 192.168.99.1 255.255.255.0

end
```

Part 4: Configure DHCP Server

Configure Static Address

Click **DHCP-SERVER**

Desktop → IP Configuration

Enter:

Setting	Value
IP Address	192.168.99.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.99.1

Configure DHCP Pools

Services → DHCP

Turn:

DHCP = ON

Pool for VLAN 10

Create:

Setting	Value
Pool Name	SALES
Default Gateway	192.168.10.1
DNS Server	8.8.8.8
Start IP Address	192.168.10.100
Subnet Mask	255.255.255.0
Maximum Users	100
Click Add	

Pool for VLAN 20

Create:

Setting	Value
Pool Name	HR
Default Gateway	192.168.20.1
DNS Server	8.8.8.8
Start IP Address	192.168.20.100
Subnet Mask	255.255.255.0
Maximum Users	100
Click Add	

Part 5: Configure DHCP Clients

Configure PC-A

Desktop → IP Configuration

Select:

DHCP

Wait several seconds.

Expected Result:

Item	Value
IP Address	192.168.10.x
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1

Configure PC-B

Desktop → IP Configuration

Select:

DHCP

Wait several seconds.

Expected Result:

Item	Value
IP Address	192.168.20.x
Subnet Mask	255.255.255.0
Default Gateway	192.168.20.1

Part 6: Verify DHCP Relay Operation

Verify PC-A

Desktop → Command Prompt

ipconfig /all

Expected:

IP Address: 192.168.10.x
Default Gateway: 192.168.10.1

Verify PC-B

ipconfig /all

Expected:

IP Address: 192.168.20.x
Default Gateway: 192.168.20.1

Verify Router Interfaces

On R1:

show ip interface brief

Expected:

G0/0.10 up up
G0/0.20 up up
G0/0.99 up up

Part 7: Verify Connectivity

Test 1: PC-A to Gateway

From PC-A:

ping 192.168.10.1

Expected:

Reply received

Test 2: PC-B to Gateway

ping 192.168.20.1

Expected:

Reply received

Test 3: PC-A to DHCP Server

ping 192.168.99.10

Expected:

Reply received

Test 4: PC-B to DHCP Server

ping 192.168.99.10

Expected:

Reply received

Test 5: Inter-VLAN Routing

From PC-A:

ping 192.168.20.100

Expected:

Reply received

From PC-B:

ping 192.168.10.100

Expected:

Reply received

Why DHCP Relay is Needed

DHCP clients send broadcasts:

DHCP Discover
255.255.255.255

Routers do not forward broadcasts.

Without:

ip helper-address

the DHCP server never receives the request.

The router converts the DHCP broadcast into a unicast packet and forwards it to the DHCP server.

This process is called:

DHCP Relay Agent

or

IP Helper Address

Troubleshooting Tips

If DHCP fails:

✓ Verify VLAN assignments

show vlan brief

✓ Verify trunk status

show interfaces trunk

✓ Verify helper address

show running-config

✓ Verify DHCP pools exist

DHCP-SERVER → Services → DHCP

✓ Verify default gateways in each pool

192.168.10.1

192.168.20.1

✓ Verify server has static IP

192.168.99.10

✓ Verify all router subinterfaces are up

show ip interface brief

What You Learned

- How DHCP broadcasts work
 - Why routers block broadcasts
 - How DHCP Relay solves the problem
 - How to configure IP Helper Address
 - How Router-on-a-Stick supports multiple VLANs
 - How centralized DHCP servers reduce administration
 - How enterprise networks distribute IP addresses across multiple VLANs
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Challenge Extensions (Optional)

Challenge 1: Add an IT VLAN

Create:

VLAN 30

192.168.30.0/24

Configure:

- New subinterface
- New DHCP pool
- New PC

Verify DHCP assignment.

Challenge 2: Add a Second DHCP Server

Create:

DHCP-SERVER2

192.168.99.20

Configure additional pools and test redundancy.

Challenge 3: Restrict Inter-VLAN Access

Create ACLs that:

- Allow all VLANs to reach the DHCP Server
 - Block HR from accessing SALES
 - Permit SALES to access SERVER VLAN only
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Challenge 4: Observe DHCP Relay in Simulation Mode

Switch Packet Tracer to:

Simulation Mode

Watch:

- DHCP Discover
- DHCP Offer
- DHCP Request
- DHCP Acknowledgment

Observe how R1 forwards DHCP traffic using the helper address.